

Project notes: High Frequency and Algorithmic Trading

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September 2026

Chapter 2: A Primer on the Microstructure of Financial Markets

2.1 Market making

Issue faced by MM:

- By providing liquidity, MMs accept one side of the trade, resulting in the MM holding the other side for an uncertain amount of time (e.g. they buy an asset but hold it for some random amount of time until someone buys it off them).
- During this holding period, the MM is exposed to the risk that the price moves against them. E.g. they buy at \$101 then by the time someone wants to buy it from them, the price has dropped to \$100.
- Given the fact MMs have no intrinsic need to hold inventory, why do they operate and exist? Grossman-Miller model shows how MMs obtain a liquidity premium that compensates them for the inventory risk. This is one reason why MMs exist.

Grossman-Miller model basically just tells us that if market participants have assets they want to sell, they cannot simply sell at the efficient price as whoever buys takes on the risk holding and potentially realising the asset decrease. Hence, if the participant wants to sell, he must sell at a price lower than the efficient price. In this market, MM and liquidity takers exist. MM will buy off the liquidity taker for discounted price. In other words, even though MMs have no intrinsic need to hold inventory, they will because they trade at favourable prices. We can interpret this as market makers setting their quotes at prices they are comfortable being long/short in. The prices they are willing to trade at have already priced in their willingness to bear the inventory risk.

- From the MM POV, compensation earned is for temporarily absorbing an unwanted position
- From liquidity taker POV, they are willing to accept a worse price as a cost of immediacy.
- The model also explains quotes not as arbitrary spreads, but as prices chosen to make inventory absorption worthwhile.
- The model includes: utility function (liquidity takers and MMs are assumed to be risk averse), fixed number of MM n in the market, liquidity takers enter the market at different times.

Trading costs:

- Distinguish between participation cost and trading cost
- Participation costs
 - Cost to actually keep models running and participate in the actual act of market making and opportunity cost of not doing something else
 - When participation cost is high, some MM would leave the market and do something else, hence there is less competition. The MM existing in the market have stronger bargaining power, therefore the cost of immediacy for the liquidity taker is high, i.e. the premium is high.
 - In reality there are many participants willing to provide liquidity (MM and other traders), hence immediacy becomes cheaper. The premium gained by MM is extremely small.

- As the cost parameter c increases in the GM model, the competition n increases, hence the premium increases.
- Trading cost
 - E.g. exchange trading fees which depend on the quantity traded, i.e. the scale of the market making being done
 - Introduce some fixed proportion of each trade as trading cost, then the GM model tells us that the trading cost shows up in the price at which the liquidity taker and MM trade.
 - The liquidity taker bears his own trading fee as well as a fraction of the market makers; the price of immediacy for the liquidity taker is even higher
 - The MM pays a fraction of his trading fee but he also trades the asset at an increasingly (compared to no trading fee model) favourable price; so the trading fee and favourable price means that it really depends on the size of the trade to tell if MM is worse off or not due to trading costs.
- Whether it be participation cost or trading cost, most of the total trading cost is borne by the liquidity taker (under Grossman-Miller)

Measuring liquidity:

- How are divergences (due to total trading costs) from the efficient prices observed in electronic exchanges?
- 2 methods: price effect, autocovariances of price changes/returns
 - They are not distinct methods but are related!!!
- Price effect: pretty much explains that the divergence from efficient prices can be interpreted as a liquidity effect.
 - Grossman-Miller model tells us the price liquidity takers and MM are willing to trade at (equation for S_1 on page 24)
 - If we decompose the traded price S_1 into fundamental value and price effect of liquidity taker asking for immediacy (page 27), we can equate the two and interpret the parameter λ as a parameter describing the liquidity of the market.
 - We see that the liquidity depends on: competition, risk tolerance, volatility.
- Autocovariance of price changes: this measure of liquidity still uses the same decomposition as above, we just compute the autocovariance of price changes
 - It's a weird concept – the second model relies on the first, so if λ is already describing liquidity, why do we need an additional model on top of our existing model?
 - Think about this: A implies B. B does not imply A, but B still tells us something about A.
 - If we don't observe everything we need to figure out A, but we observe B, we can still infer something about A.

Market making using limit orders:

- At a high level, this subsection deals with how a market maker quotes their bid and asks to maximise their profits.
- Say the MM places a bid and ask at δ^- and δ^+ away from the mid price, i.e. bid = $S_t - \delta^-$ and ask = $S_t + \delta^+$.
- The question is then: how does the MM choose this distance away from the mid price?
- Intuitively, the MM wants to maximise profits.
 - If they quote too aggressively (that is, price too far away from the mid price), their quotes aren't competitive. i.e. the probability that the LO is executed is low.

- On the other hand, pricing too conservatively will increase the probability of the LO being executed but decrease the profit per trade.
- MM wants to balance this trade-off.
- MM quotes their bid and asks such that their expected profit per trade is maximised (this optimisation problem balances the above trade-off).

Thoughts: even under this simple model, the optimisation problem that arises is pretty complex. Transitioning from a static to dynamic market model will make things exponentially harder. Simpler models are still useful in providing intuition. Maybe for my project, I can just isolate one mechanism and study how they interact. It's more realistic because I don't have the ability or resources to build a fully dynamic (or anything close) model.

2.2 Trading on an Informational Advantage

Informational advantage, price impact:

- If a market participant believes they have an informational advantage, they would obviously want to trade on that.
- One thing they have to take into account is the price impact; the impact on the asset's price as a result of trading on that information.
- Others will observe their MO (this is a naive interpretation as in reality participants cannot tell the reason behind MO) and adjust their prices accordingly. Hence, aggressively trading to exploit their information quickly can move against them as market participants adjust their prices to reflect the imbalance of information.

Kyle (1985) model:

- Similar to Grossman-Miller where we have liquidity takers and market makers.
- The difference is that these liquidity takers are insensitive to prices; they will always trade no matter what. The market makers are now risk neutral instead of risk averse, hence they don't require premium to hold inventory.
- Introduce an insider who has an informational advantage. In this scenario, this means that they know the fundamental value of the asset v (compare this to the price it's traded at, S).
- Strategic interaction between market participants: taking into account rationality and optimal behaviour, each and every participant takes into account each other's behaviour into how they will behave. i.e. market makers set their prices as a function of observed net order flow (a function of order flow arising from insider and liquidity taker), the insider takes into account the price impact of trading on this informational advantage.
 - The insider cannot simply trade an arbitrarily large amount on their information, because their own order moves price against them.
 - The market makers know that order flow may contain informed trading, so they condition their quotes/prices on observed net order flow.
- The liquidity takers act as camouflage for the insider. Without liquidity takers, MM will know any incoming MO arises from insiders, hence their prices constantly anticipate this, resulting in insiders not being able to exploit their informational advantage at all.

Traded price and distinction between Grossman-Miller:

- The traded price in the Grossman-Miller model is set by the MM and includes an explicit risk premia to account for the random future price movements. MMs do this because in this model, they are risk-averse.
- Kyle's 1985 model assumes MMs are risk-neutral. They do not add the same risk premia. However, they do know that insiders exist within the market they operate in, and add a "risk premia" for this (not exactly a risk premia but a loading in a sense).

- MMs observe net order flow consisting of uninformed liquidity trading and informed trading by the insider. Although the MM doesn't know which one is the insider, he observes the net order flow which lets him update his estimate on what the fundamental value of the asset may be.
- Heuristically, the MM averages out the informational risk onto every quote. But because it is averaged, there is some amount that the insider can trade such that he can still make a profit
 - Mathematically, the MM sets price using the expected fundamental value conditional on observed order flow.

2.3 Market Making with an Informational Disadvantage

The previous section is similar but was insider-centered. We now look at the same scenario but from the lens of a MM.

Glosten & Milgrom (1985) model:

- Model setup is very similar to sequential model in Hasbrouck
- Two possible fundamental values: V_H, V_L (the multiple insiders in the market know the true value)
- Proportion of insiders: α
- Insider/informed trading behaviour: depend on whether they can buy for a if $a < V_H$ and sell for b if $b > V_L$. Unconditional probability that $v = V_H$ is p .
- Liquidity taker/uninformed trading behaviour: buy and sell with equal probability, independent of the true value of the asset
- MM problem is to choose their ask a and bid b .
- Competition drives the expected profits of MM to zero, and with the probabilities of buy and sell and proportion of informed/uninformed traders, we can find a and b by setting expected profits = 0.

Extensions of the model:

- The model that is set up is static through time. We can vary time in a similar way to how Hasbrouck did; the bid and ask prices are updated using the accumulation of public information from trade captured by the filtration $\mathcal{F}$.
- We can also make liquidity traders price-sensitive, that is, a proportion of liquidity traders do not trade.
- In the first model, all liquidity traders trade, either buying or selling with probability p and $1 - p$. Now we consider a proportion of total liquidity traders buying and selling. This means that there will be a smaller population of liquidity traders. Intuitively there is less noise in the market to camouflage the insiders. MM increases their spread to price the increased risk of adverse selection.
- This is a feedback effect: wider spread $\rightarrow$ fewer liquidity traders $\rightarrow$ less noise/camouflage $\rightarrow$ executed order flow is more informative $\rightarrow$ greater adverse-selection concern $\rightarrow$ wider spread $\rightarrow \dots$

Chapter 12: Order Imbalance

12.1 Intraday Features

Measuring order imbalance:

- We define LO imbalance as the ratio of quoted volume imbalance to total quoted volume

$$\rho_t = \frac{V_t^b - V_t^a}{V_t^b + V_t^a}$$

- Where the numerator is the difference in bid and ask LO volume and the denominator is the total LO volume

- LO volume can be measured in many ways: at-the-touch (i.e. take the best bid and ask; this is very local), all existing LO, or all existing LO within a certain range of the midprice.
- Any measure chosen is fine as long as it is fit for purpose
- When we model order imbalance we end up with a time series. We can split the ratio into bins to represent different market regimes based on varying fixed intervals of LO imbalance. E.g. regime 1, 2, 3, 4, 5 corresponds to when LO imbalance $\rho_t \in [-1, -0.6), (-0.6, -0.2), \dots, (0.6, 1]$ respectively.
 - From there, we can compute $\mathbb{P}(\text{market in regime } k \mid \text{buy MO})$, $\mathbb{P}(\text{market in regime } k \mid \text{sell MO})$ for $k = 1, 2, 3, 4, 5$.

Discrete-time Markov chains:

- We can generalise and let $Z_t = 1, \dots, K$ to denote the order imbalance regime observed at time t
 - For example, at time $t = 3$ we want to find the probability of being in each regime. We then use our observations of order imbalance at $t = 0, 1, 2$ to see which regime we were in at time 0, 1, 2 and methodically assign a predictive probability (it can be done many ways).
- Let the transition matrix be $\mathbf{A}$. We can estimate $\mathbf{A}$ using:
 1. MLE: probability from state i to state $j = \# \text{ transitions from state } i \text{ to state } j / \text{total transitions}$.
Derivation on page 298, problem is maximising likelihood constrained to sum of probability of transition from fixed i to $k = 1$, i.e. $\sum_{j=1}^K (A_{ij}) = 1 \rightarrow \text{Lagrange multiplier method}$.
 2. Estimation from data: vocab used “stickiest”, “transition pressure”

Continuous-time Markov chains:

- We can estimate continuous-time Markov chain generator matrix $\mathbf{B}$ using discrete-time Markov chain transition matrix $\mathbf{A}$
 - i.e. we transform the discrete-time into continuous-time.
- The rows in $\mathbf{B}$ sum to 0: we interpret $\mathbf{B}$ as a flow of transition rates.
 - B_{ij} is the instantaneous rate of jumping from state i to state j
 - The diagonals are negative as it represents the probability mass flowing out of the current state.

Thoughts: we can partition order imbalance into equally spaced intervals and make them represent regimes under different order imbalances. These order imbalances may arise from probability of buy and sell that change over time (perhaps they are random variables from a distribution that economically makes sense or just uniformly random for now), but the real question is how are we able to detect when the regime has changed? Naively, something like conditional expectation: we want expected regime at next trade given all information (modelling choice is to determine how much memory the model has).

- Or better yet since we are modelling something uncertain, instead of a point estimate we can find the full probability vector (conditioned on all information) over next regimes to get a distribution.

Modelling MO arrivals:

- Overall idea:
 - Assume that, conditional on being within a given regime k , the buy and sell orders arrive independently and are modelled as Poisson processes and are denoted as λ_k^+ and λ_k^- respectively.
 - So each regime $k = 1, \dots, K$ will have corresponding buy and sell order processes which are independent from each other.
- Switching times, epochs:

- Denote the switching times (this is the time at which the Markov chain changes regimes): $0 = \tau_0 < \tau_1 < \dots, < \tau_N < \tau_{N+1} = T$.
- Divide $[0, T]$ into $N + 1$ epochs. Then, within epoch r , $[\tau_r, \tau_{r+1})$, the regime is constant. i.e. within an epoch the Markov chain does not switch states/regimes.
- We now denote the arrival times of buy and sell MOs within the r -th epoch as $\{b_{r,1}, \dots, b_{r,m_r}\}$ and $\{s_{r,1}, \dots, s_{r,n_r}\}$ respectively. i.e. in the r -th epoch there are m_r buys and n_r sells.

Modelling price jumps:

- Determining the distribution of price changes conditional on the arrivals of buy and sell MOs and order imbalance regime prior to arrival of that MO
 - i.e. predict a price change distribution for the very next MO using “all information”